

Vayu: Roadmap and Deployment Path

Supporting document, SIH 2026, ByteBrigade. Planned work, ordered by dependency. Items are plans, not delivered features.

1. From prototype to a defensible index

Step	Work	Why it comes in this order
1. Re-base	Set each route's reference price from a surveyed base period (a complete run) so the base day reads exactly 100. Recompute history from stored route detail.	Without it, the index level cannot be interpreted. Everything else builds on a valid base.
2. Real weights	Replace the illustrative traffic masses with DGCA sector-wise passenger traffic; regenerate the basket with the existing script; revise on a fixed cycle.	Weights determine which routes move the headline number.
3. Matched-route change	Publish day-on-day change computed on routes present on both days, alongside the headline index.	Separates fare movement from changes in route coverage.
4. More sources	Enable the IndiGo and Air India parsers in the scheduled run, then SpiceJet and Akasa. Cross-check sources against each other.	Reduces dependence on one aggregator.
5. Advance-purchase windows	Add further windows beyond the current 21 days ahead.	Statistical offices collect airfares at several lead times.
6. Cabin and fare-class filter	Restrict to economy fare families at parse time.	Keeps the fare basket consistent.
7. Hardening	Authenticate and rate-limit the write endpoints; alerting when coverage drops; refuse to publish an index when no route was scraped.	Needed before public use.

2. Deployment path

Stage	Scope	Exit criterion
Prototype (current)	Single source, 325-route basket, daily run, public dashboard	Pipeline runs unattended for consecutive days
Pilot	Re-based index, DGCA weights, matched-route change, two or more sources	Index reviewed against published airfare statistics and explained
Hand-over	Documented method, monitoring, authenticated access for analysts	Statistical agency can run and audit it